Accident Data Analysis Report

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1. Introduction

Analysing real-world data regarding road traffic accidents in Great Britain of the year 2020, this report answers set questions regarding accident changes, circumstances and zones. structured way, and the results are then searched for temporal, spatial, and context factors to determine accident hotspots, causes and future trends. Sophisticated tools such as Data clustering, social network analysis, and time series analysis are thereby used to arrive at suggestions on how road safety can be enhanced.

2. Analysis

There are 4 significant tables in the road accident database: 'accident,' 'casualty,' vehicle,' and 'lsoa', which provides details on the location of an accident, its seriousness, vehicles and people involved and their ethnicity respectively. Data for 2020 is queried using SQL and imported in the form of Data Frames with the relevant records being linked (LSOAs to accidents, for example). Duplicates are resolved to remove any problems in data quality and missing values are replaced with either a placeholder or imputations(Arena and Pau, 2020). During data pre-processing for analysis, the data is cleaned for intrinsic analysis; future enhancements could be on data imputation from its domains and data pre-processing inconsistency from the databases for increased credibility.

➢ When Do Accidents Happen?

i. General Trends

- **Day of the Week:**There is a high number of accidents, and more particularly 14, 889 on Fridays requiring more than normal attention compared to other days of the week. This pattern is attributed to increase in traffic and after working many hours during the week, together with social or other pleasurable activities. Congestion on Fridays seems to contribute a lot to increased accident incidences as motorist scrambles to settle their business or get back home. Wednesday with 13,564 and Thursday with 14,056 means that during mid of the week, hence in middle of the week, there exist some risks with traffic.

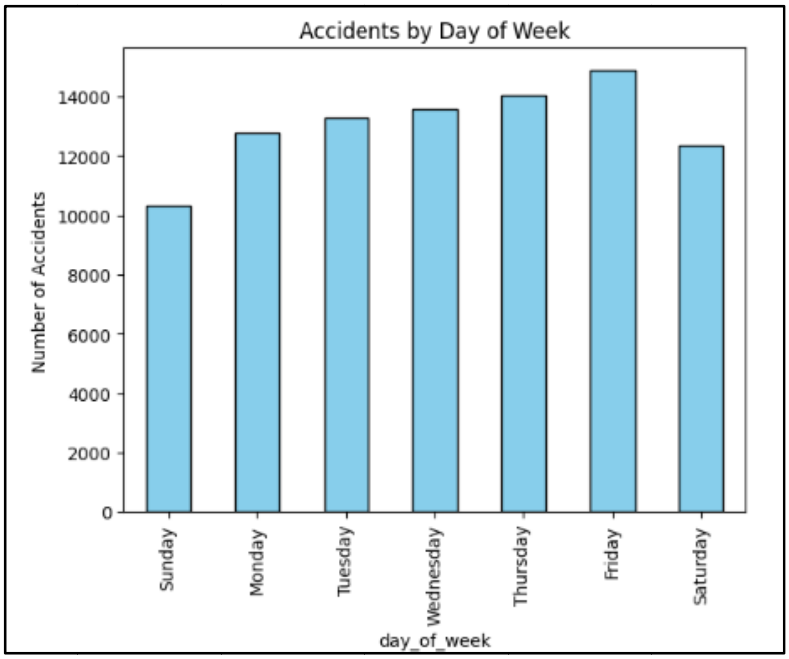

Figure 1: Significant days of week

- **Hour of the Day:**Accidents have unique temporal distribution; the highest occurrence is during the evening rush hours. The hour of 17:00 (7,813 accidents), which is determined to be the most dangerous, and as the people commute from work, and travel in the evening, increase. Secondary peaks occur at 16:00 (7,381 accidents) and between 15:00 and 18:00, which is attributed to school and work-related traffic congestion.

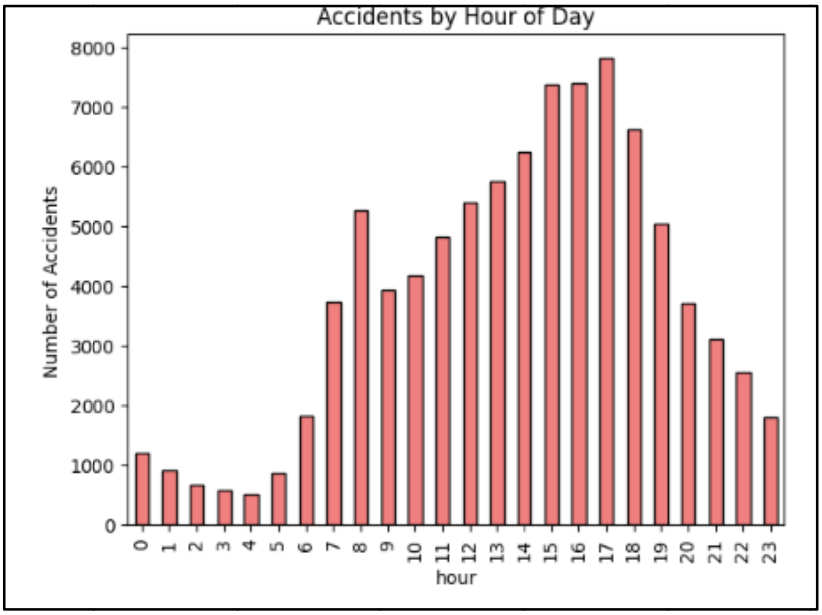

Figure 2: Significant hours of day

Sundays had the least number of accidents (10,315) since cars are few on the road while Saturdays recorded (12,336) attributed to recreational and social trips. Early mornings

(00:00–06:00) observe few accidents, and a slight rise at 06:00 (1,830) considering that it is around the rush hour in the morning. The frequency of accident occurrence is highest in the evening and then most probably decreases.

ii. Motorcycle Accidents

- **By Day:**According to the days of the week, Fridays record the highest motorcycle accidents at 2119, Thursdays at 2002 and Wednesdays at 1841, as observed from the week commuter traffic. Monday (1,703) so has the least count because there is less traffic, but Saturdays (1,830) increase again probably because of non-commercial use.

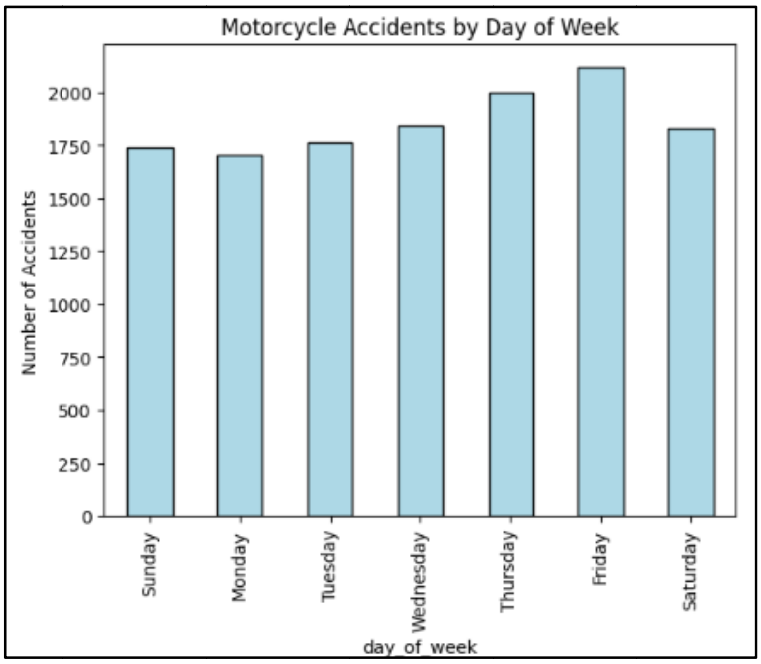

Figure 3: Motorcycle accidents by day of week

- **By Hour:** Evening rush hours peak at 17:00 (1288 accidents), the results obtained correspond to the traffic density and the fatigue. Another rise is observed at a period from 12:00 to 15:00 to reach its maximum at 15:00 with 1038 which may refer to delivery or leisure riders. Minimal accidents occur during early morning hours (00:00–06:00), accidents gradually rise from 06:00 (243) in the morning to evening.

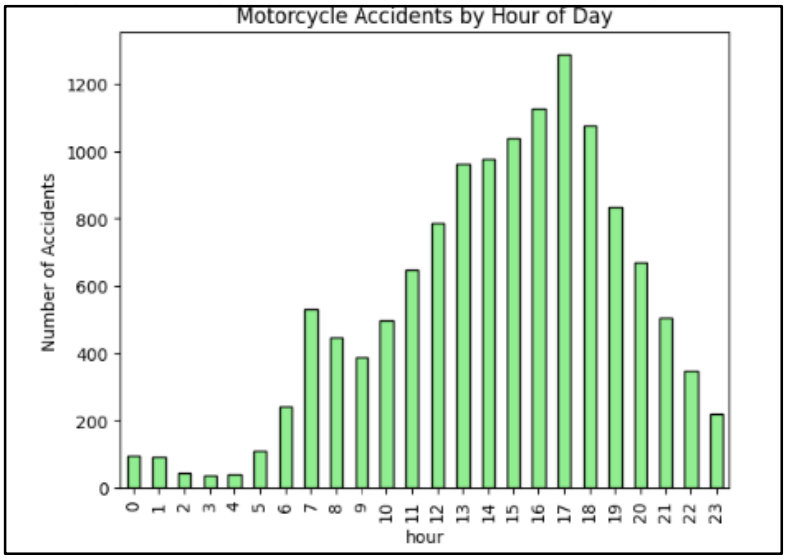

Figure 4: Motorcycle accidents by hour of week

iii. Pedestrian Accidents

- **By Day:**Fri (2,543) has the highest pedestrian risks than any other days in a week due to weekday activities while Thurs (2,366) and Tues (2,267). Sundays remain the least with 1,242 while Saturdays have a slight increase due to the weekend with 1,878.

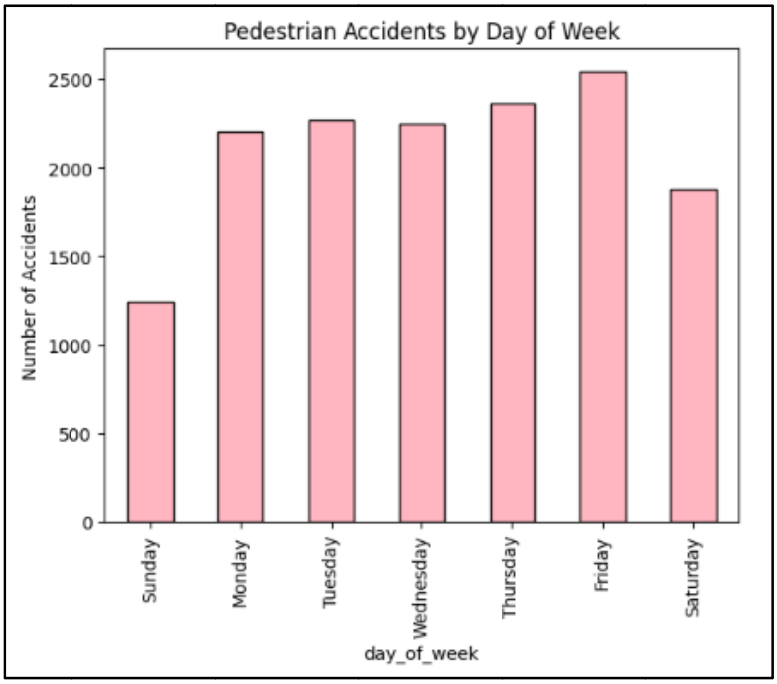

Figure 5: Pedestrian accidents by day of week

- **By Hour:**The largest number of accidents is recorded at 15:00 (1,672) associated with schools' dismissal, then followed by 16:00 (1,323). Mornings peak at 08:00 (1,060) during drop-offs. Late-night hours (00:00–06:00), except for 04:00 which had 50,

drivers report few accidents mainly influenced by driving under the influence or high risk from poor visibility.

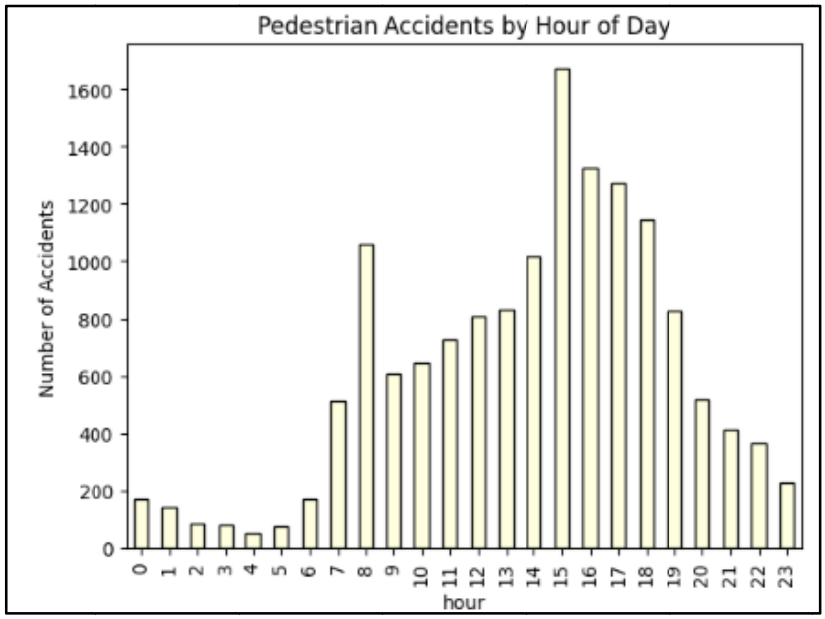

Figure 6: Pedestrian accidents by hour of week

➢ Where Do Accidents Happen?

i. Regional Clustering

- **Using K-Means for Spatial Analysis:**The clustering analysis used the K-Means algorithm to find the distribution of the accident pattern in Kingston upon Hull, Beverley, and the East Riding of Yorkshire. To control for bias related to driver origin, the dataset was limited to these counties using their district codes (220, 228, and 231), focusing the analysis on urban and suburban accident zones(Sinaga and Yang, 2020). Where latitude/longitude values were not available, they were replaced by 'unknown' value to achieve clear input clustering.

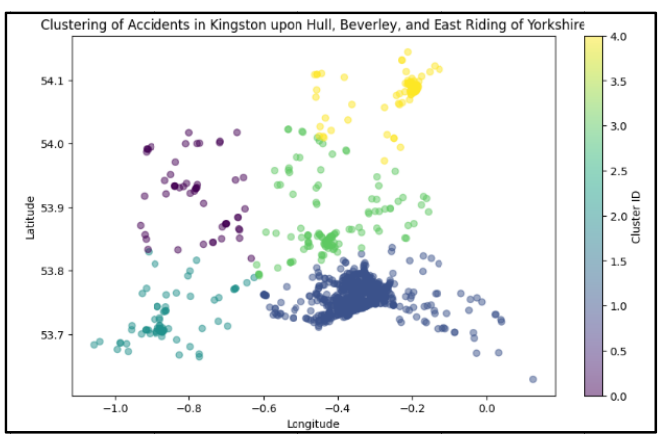

Figure 7: Clustering of accidents in Kingston upon hull, Beverley and east riding of Yorkshire

The provided dataset includes the latitude and longitude features that were fitted and scaled by using the StandardScaler. By doing so, both of the coordinates played equal role in the clustering algorithm and did not lead to distorted results due to different magnitude of the two dimensions. The data is further divided into five clusters (n_clusters=5) by checking the density and accident spatial distribution. Record numbers were labelled as cluster ID and visual representation of the accident categories developed for the analysis of clusters.

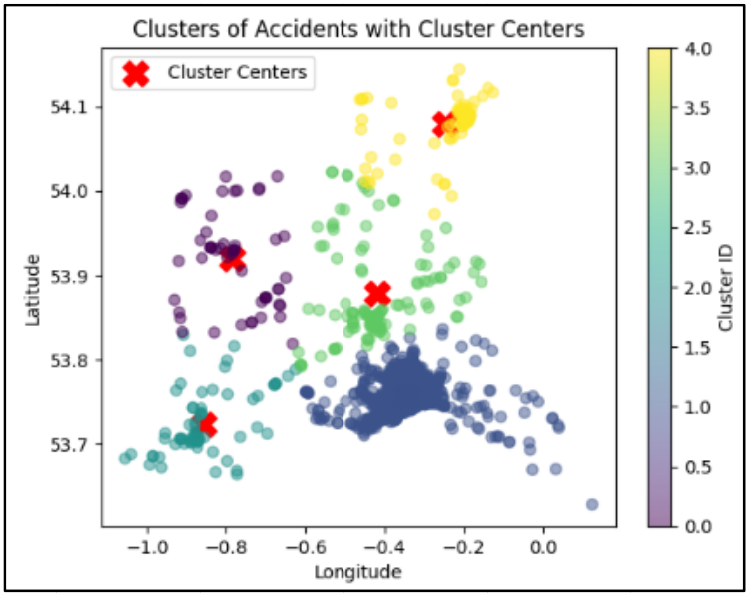

Figure 8: Clusters of accidents with cluster centers

- **Cluster Insights:**This resulted in clusters comprising of areas that contain elements of both urban and suburban settings as well as accident-prone areas within highways. Places such as Kingston upon Hull had high traffic density concentrations because of crowded traffic flow, complicated number of carriageways, frequent conflicts between the pedestrian and vehicle domains and so on(Gormley, Murphy and Raftery, 2023).

ii. Top LSOAs in Hull

- **High-Accident LSOAs:**To determine which specific locations had the highest accident rates early 2020 in Hull city, the Local Super Output Areas (LSOAs) were utilized. Specifically, local_authority_district was the column that the data was further refined to include only Kingston upon Hull (district code 228). Finally, this dataset was subset to the first quarter, January to March 2020 accidents.

The accidents were grouped by the LSOA codes, and the top three LSOAs with the highest counts were identified:

- **E01012817:**Mentioned 10 accidents, which means that it is identified as the most 'accident prone' LSOA in the dataset.
- **E01012848 and E01012889:**Each recorded seven accidents and, therefore, featured high accident density rate.

These LSOAs are indicative of the specific high-risk areas which need safety intervention at the earliest.

➢ Under What Conditions Do Accidents Happen?

i. Association Rules

- **Using the Apriori Algorithm to Highlight Patterns:**The use of Apriori algorithm enabled the determination of Affected Factors for accident severity concerning the environmental and situational factors(Zhang and Zhang, 2023). The features selected were therefore coded on binary form with their values from the data set including the road type, weather, light, speed limit and road surface condition.
- **Example Key Findings:**
  - **Weather and Light Conditions:**Some of the factors include visibility especially during certain weather type (rain, fog), or certain time of the day or night, (dusk, night) that have adverse effect on the road condition and subsequently raises the accident rates. For example:
    - The predictor variable of 'light conditions, road surface and type of road' had a confidence value of 82.9% for minor accidents which shows that visibility plays a crucial role in averting the accident.
    - One of the significant rules that admit road surface and condition, road type, and weather factors combined had the lift value of 1.056, which underlines how these factors influence the accident severity in comparison with the other factors.

```
Rules Summary:
12
{road_type} -> {accident_severity_3}
0.051393
0.839964
1.072088

74
{road_surface_conditions, road_type} -> {accident_severity_3}
0.035987
0.834265
1.064814

71
{light_conditions, road_type} -> {accident_severity_3}
0.035976
0.834011
1.064490

164
{light_conditions, road_surface_conditions, road_type} -> {accident_severity_3}
0.027939
0.829427
1.058639

111
{road_surface_conditions, road_type, weather_conditions} -> {accident_severity_3}
0.033378
0.827624
1.056337
```

Figure 9: Top 5 Rules summary by apriori algorithm

ii. Social Network Analysis

Road traffic accidents were modelled as a network, where each equivalent accident is a node in a graph, created using NetworkX. Edges were established on the basis of the presence of vehicles or living in the adjacent Lower Super Output Area (LSOA). This approach made it possible to get deep understanding about the dependencies between accidents and their connections(Prabhakar and Anbarasi, 2021).

- **Network Characteristics**:
  - The graph is consisted of 91,210 nodes and 7,734,717 edges, which suggest that the data set has high density of connectivity.
  - According to the value of density estimated at 0.0019, it means that the network is quite spread out but contains a high level of modularity.
  - **Average degree**: On average there were 169.6 connections between each node, which means that accidents are either connected by the involvement of similar vehicles or by location.

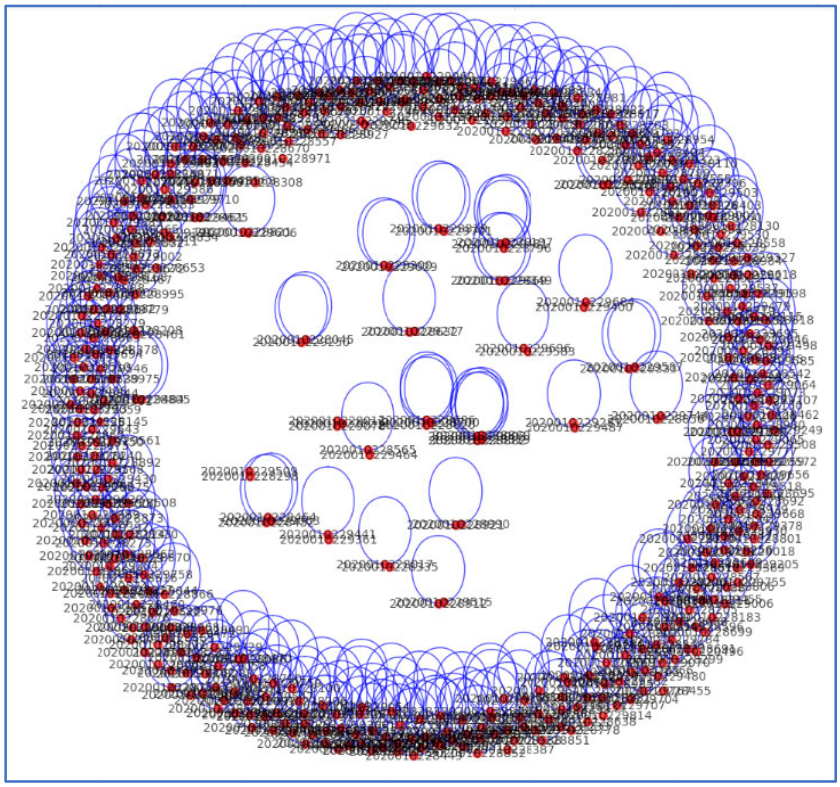

Figure 10: Sample of social network graph

For the creation and analysis of graph, NetworkX of Python was employed with an emphasis on shared vehicle and LSOAs. A subset visualization (500 nodes) provided evidence of densely connected groups and in many cases these groups corresponded to accident-prone urban areas.

iii. Edge Betweenness Centrality Analysis

Edge betweenness centrality was computed to determine criticality of connections in the network and to highlight edges that connect different clusters.

- **Key Analysis**:
  - Edge betweenness centrality determines the shortest paths that cross the edge and hence identifies "bridge" edges that are important for the connection of the network.
  - **Skewed Distribution**: Most edge nodes with low centrality scores suggested that they mainly help interconnect the clusters. There were only a few high centrality edges that would effectively connect the various clusters.
  - Histogram Visualization was used to identify the distribution of the centrality values; results showed that the network was indeed modular.

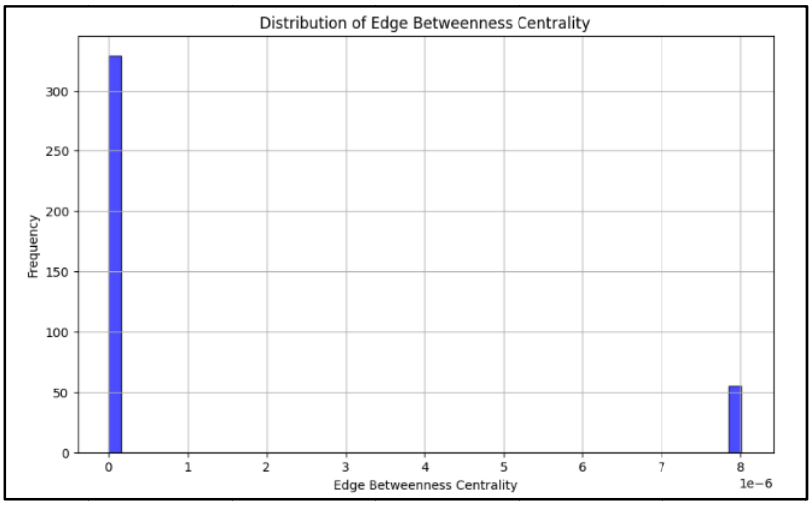

Figure 11: Distribution of edge centrality

In this case, low-centrality edges are actually dense urban connections, while high-centrality edges are important road connection between accident clusters that require intervention. This localised accident patterns reveal that the network is modular and has loose connections between clusters.

iv. Community Detection Using Girvan-Newman and Louvain Algorithms

To analyse the patterns of the nodes interconnected in the accident map, two community detection algorithms, namely, Girvan Newmann and Louvain were utilized.

- **Techniques and Libraries**:
  - **Girvan-Newman Algorithm**:

- Recurrently eliminated connections with the highest betweenness centrality to identify hierarchical structures within the communities.
- Led to 453 clusters, where smaller and scattered clusters were detected because of the elimination of edges.

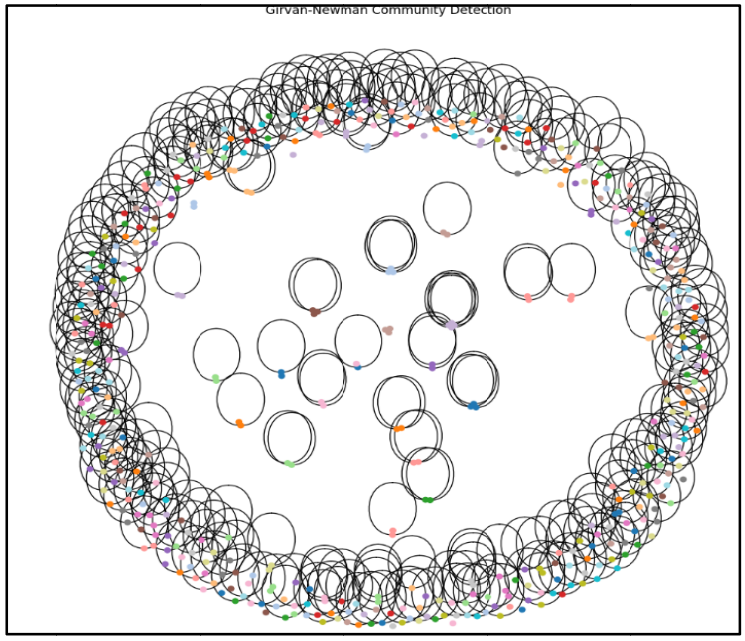

Figure 12: Girvan Newman Community Detection from sample network

- **Louvain Algorithm**:
  - The modularity scores optimized to divide the graph and emphasize internal connections within the cluster.
  - Identified 452 clusters; the communities were slightly larger compared to Girvan Newman.

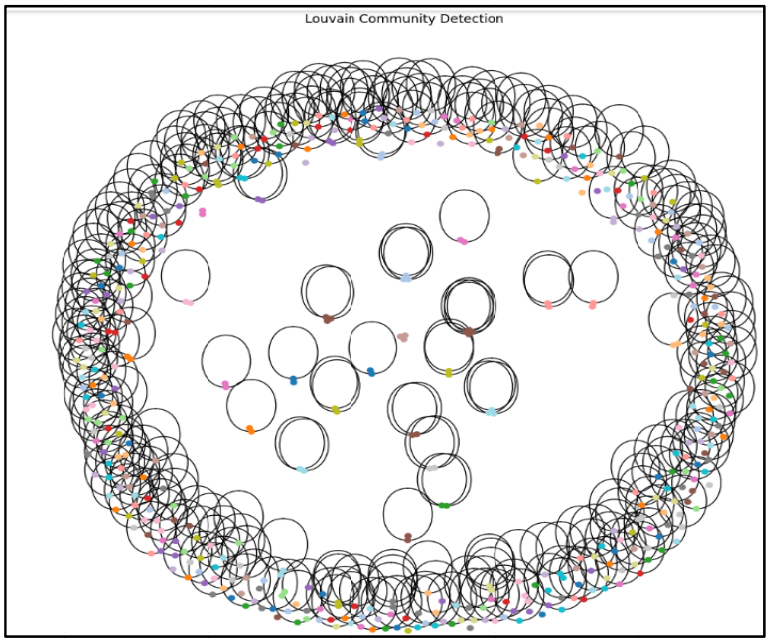

Figure 13: Louvain Community Detection from sample network

Girvan-Newman identified more fine-grained clusters, Louvain identified bigger, more coherent modular structures, both in line with urban accident hotspots.Community identification reaffirms separation of different areas as accident-prone regions. Girvan-Newman suits a detailed level; Louvain shines for scalability to provide focused city planning and traffic control.

3. Predictions

i. **Weekly Predictions on data from 2017 to 2019 (based on three policing areas by filtering the data using the "police_force" column, i.e., Warwickshire, Derbyshire, Nottinghamshire)**:

- **Warwickshire**: A stable range of weekly accidents ranging from 10 to 30. It can thus be ascertained that the forecast made by the model is accurate given that the forecasting error has a Mean Absolute Error of 5.65.

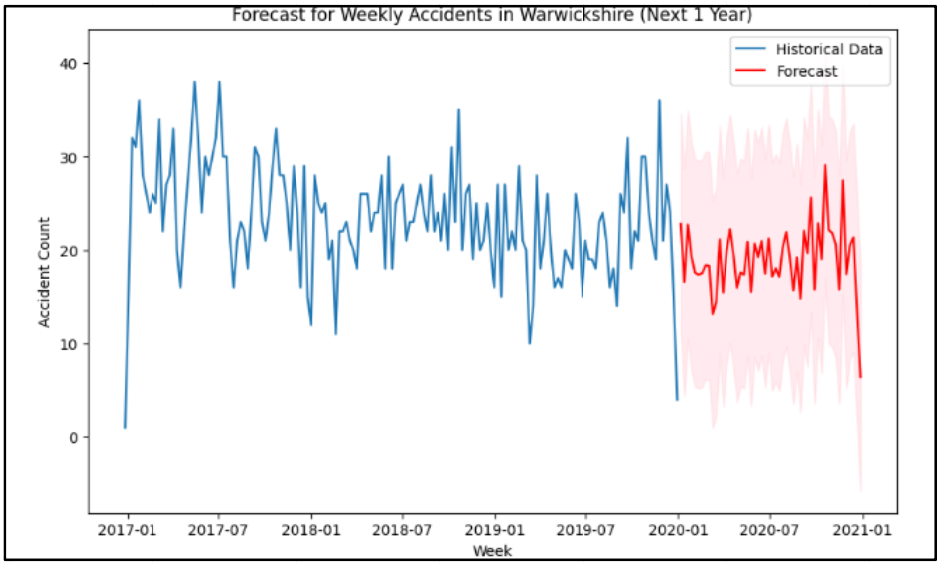

Figure 14: Forecast for weekly accidents in Warwickshirefor next 1 year

- **Derbyshire**: More variation, with 20-40 accidents weekly. The model was relatively good (MAE = 8.71).

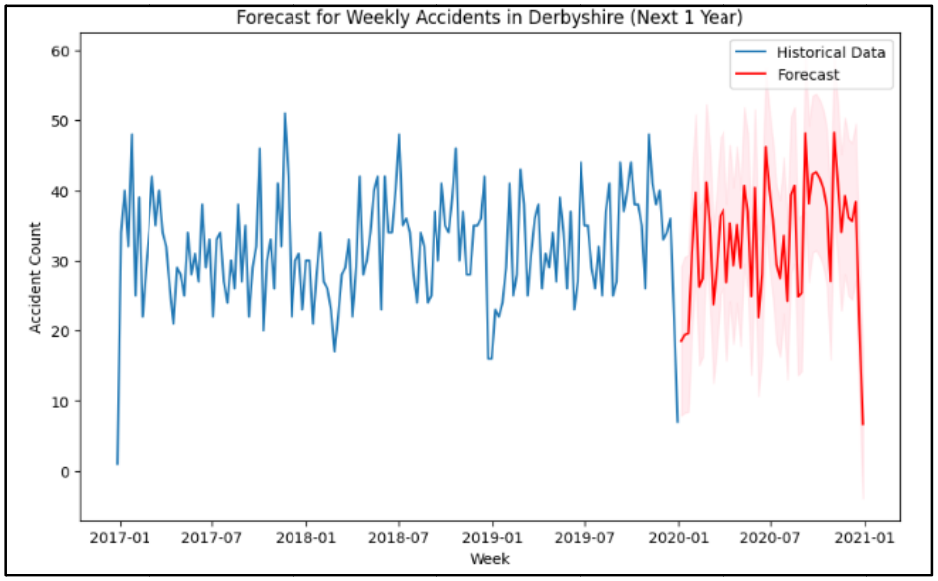

Figure 15: Forecast for weekly accidents in Derbyshirefor next 1 year

- **Nottinghamshire**: More variation, 30-50 accidents in a week. Seasonal trends are most influential on accuracy (MAE = 9.21).

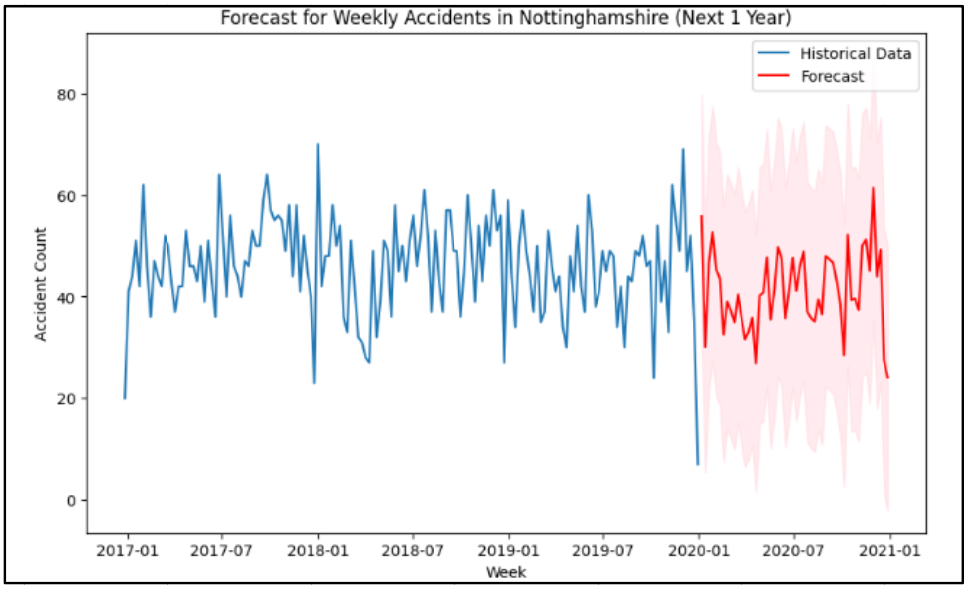

Figure 16: Forecast for weekly accidents in Nottinghamshirefor next 1 year

The variation of the data and its seasonal pattern are important to the prediction. In areas where long-term trends provide a clear picture, advanced predictive data is superior; a high variability, such as in specific geographic locations may necessitate further improvements.

ii. Daily Predictions for Hull's High-Accident LSOAs:

- **E01012817:** The constant fluctuation over the day with occasional high lapses. Ability to discover wider confidence intervals in forecasts is indicative of higher levels of uncertainty**.**

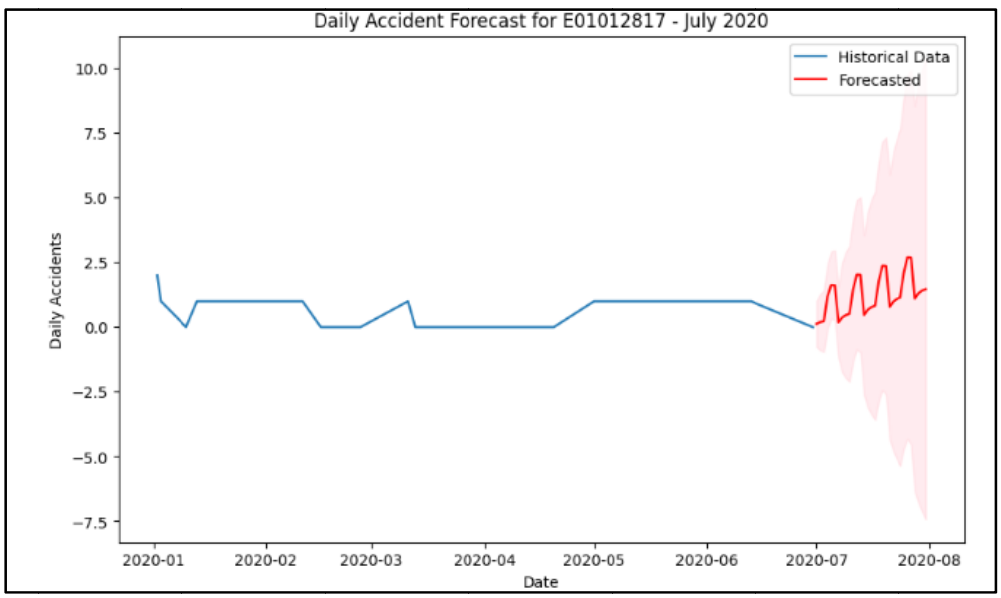

Figure 17: Daily accident forecast for E01012817 – July 2020

- **E01012848 & E01012889:** Lower accident risk levels, with reliable and constant trends but unpredictable projections owing to the scarcity of data**.**

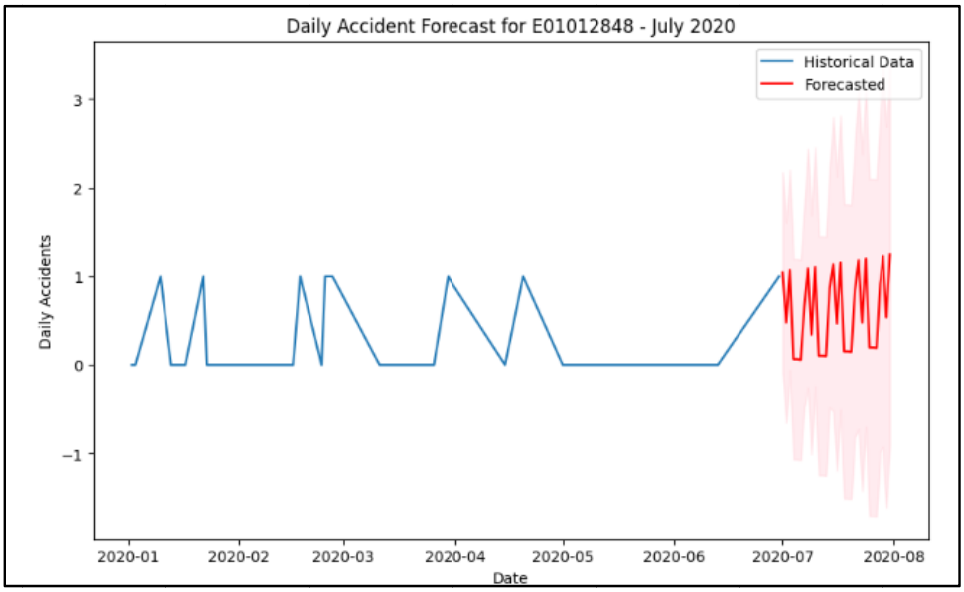

Figure 18: Daily accident forecast for E01012848– July 2020

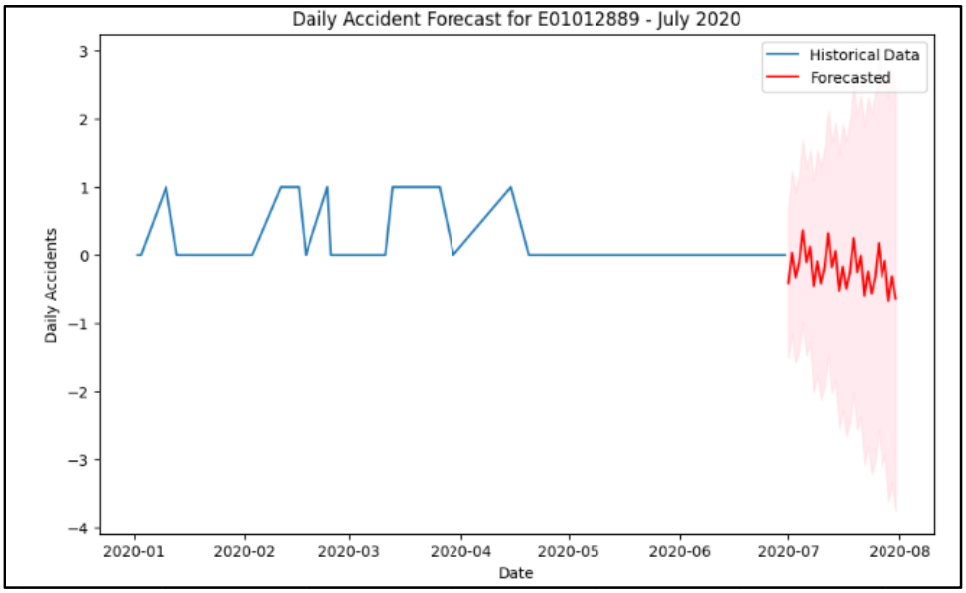

Figure 19: Daily accident forecast for E01012889– July 2020

Frequency of data is another factor, which indicates that high-frequency data leads to better forecasts than low frequency data input. Emergency measures should be taken for LSOAs with many accidents, such as E01012817, although more data might improve prediction for less risky zones.

4. Recommendations

- **Target Urban Hotspots**:

- LSOA's such as E01012817 in Hull and similar should be given preference. These measures would include installing speed humps, cameras or other similar types of traffic calming devices as well erect more signs and increase illumination on such areas.

- **Time-Sensitive Interventions**:
  - Increase enforcement during high-risk times:
    - which is morning (08:00) and evening (15:00–18:00)
  - Deploy school crossing guards during peak pedestrian hours (**15:00–16:00**).
- **Environment-Specific Measures**:
  - Fix problems related to weather and lighting. There should be reflective markings and anti skid on all black spots areas that are usually slippery specially during wintertime.

- **Public Awareness Campaigns**:
  - Counsel motorcyclists and drivers more on how to avoid accidents during rush hour. Promote pedestrian safety, more so for children and especially school zones.
- **Data-Driven Policy Making**:
  - Collect more historical data for improving the forecast model.
  - Implement cluster and social network approach for correct resource allocation.

5. References

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